

Internal Assessment (IA) Portal Development Report

1. Problem Statement

In a rapidly evolving educational environment, the need for a structured and transparent assessment management system is more crucial than ever. Current manual methods for handling assignments, tracking submissions, and organizing courses create significant challenges for both teachers and students, resulting in delays, lost assignments, and limited visibility into assessment progress.

The Internal Assessment (IA) Portal seeks to address these issues by providing a centralized platform where teachers can seamlessly create and manage assignments and quizzes, and students can access, complete, and submit their work. The portal is designed to bring real-time submission tracking, clear communication of deadlines, and structured course management, ensuring that both teachers and students are informed and engaged throughout the assessment process.

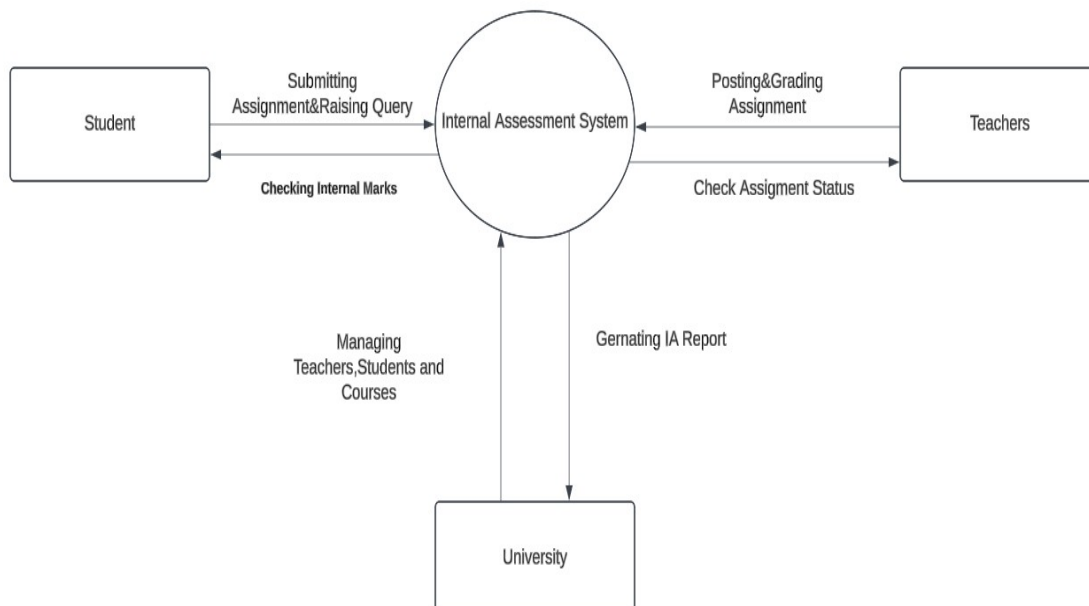
Along with that it provide interface for the university/college to create course,student,teachers and assign courses to them.

2. Process Model

We use a mix of **Incremental Process Model** and **Evolutionary Process Model** for our project which mean dividing the project into smaller independent part and start building it independently then start a new model and developing that

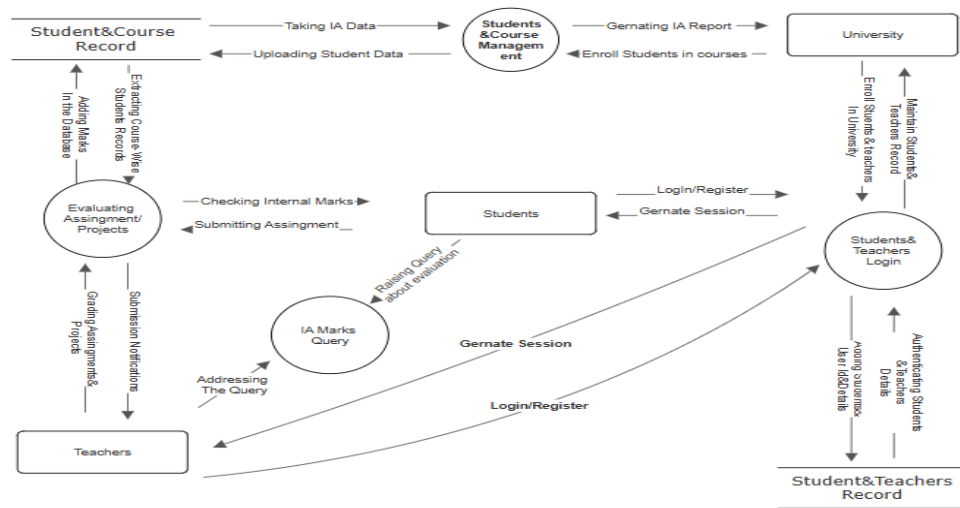
2. Data Flow Diagram

- Zero Level DFD



- 1 Level DFD

1 Level DFD



DATA DICTIONARY

Student:

- studentID:char+char+char+digit+digit+digit&(required and unique)
- Name:String (required)
- Email:string,(required , unique)
- Password:HashValue(Default:"NOT-SET")
- DOB:YYYY-MM-DD
- Gender:"MALE" OR "FEMALE"

Teacher

- teacherID:char+char+char+digit+digit+digit&(required and unique)
- Name:String (required)
- Email:string,(required , unique)
- Password:HashValue(Default:"NOT-SET")
- DOB:YYYY-MM-DD
- Gender:"MALE" OR "FEMALE"

Course

- courseID:char+char+char+digit+digit+digit&(required and unique)
- Name:String(Required)

courseAllot:

- courseID:char+char+char+digit+digit+digit&(required and unique)
- userID: char+char+char+digit+digit+digit

- Role:Instructor or Student

Assignment

- AssignmentID:'A'+courseID+:+digit+digit+digit
- Name:String
- createdBY:TeacherID
- subjectCode:courseID
- endDate:Date;
- Max Marks:Integer
- Content:String
- assignment Path:path

IA Report: Excel or PDF File

Submission

- submissionID:'S'+courseID+:+digit+digit+digit;
- assignmentID:assignmentid
- submissionerID:studentID
- submissionDate:Date+time;
- Marks:integer
- submission content:String
- submissionPath:path of submitted file

SRS Document

1. Introduction

1.1 Purpose

The purpose of this document is to provide a comprehensive description of the **Internal Assessment (IA) Portal**. This portal is designed for academic institutions to facilitate the process of creating, managing, and submitting assessments (assignments and other resources) within a centralized platform. The portal aims to streamline communication between teachers and students, enabling efficient submission tracking and grading.

1.2 Scope

The IA Portal will:

- Allow teachers to create assignments and resources for students.
- Enable students to view assigned assessments and submit their work.
- Manual grading by teachers and generating automatic IA Report.
- Provide a dashboard for teachers and students to view the status of assessments.
- Allow university to manage course enrollments, teachers and Students.
- Maintain a secure database of assignments, submissions, and results.

1.3 Definitions, Acronyms, and Abbreviations

- **IA:** Internal Assessment
- **Assignment:** Academic work assigned to students for completion.
- **Quiz:** A test with questions for assessing student knowledge, which can be graded automatically.
- **Course:** A subject or class that students are enrolled in and linked to assessments.
- **Teacher:** A faculty member responsible for creating and managing assessments.
- **Student:** A user of the portal who completes and submits assignments and quizzes.

1.4 References

- Academic assessment policies.
- Database schema for the IA Portal (MongoDB or equivalent).
- Data flow diagrams for system architecture.

2. Overall Description

2.1 Product Perspective

The IA Portal is a web-based application intended to be used by academic institutions. It will integrate with existing student information systems to provide a smooth experience in managing assessments. Teachers can assign tasks, and students can view and submit them within the portal. The portal will store assignment and submission data, allowing for easy access to historical information and records.

2.2 Product Functions

Teacher Functions:

- Create Assignment and grade them.
- Set deadlines and maximum marks for each assessment.
- View and track student submissions.
- Grade submissions manually and update the system with scores.

Student Functions:

- View assigned assessments for enrolled courses.
- Submit assignments and quizzes before deadlines.
- View submission status and grades once graded.

University Functions:

- Manage teacher and student accounts.
- Oversee course assignments and enrollments.
- Maintain system logs and backup data.

2.3 User Characteristics

- Teachers: Experienced with managing student assessments and familiar with using web applications.
- Students: Regularly use web applications, able to upload files, and raise query
- University: Experienced with managing users and troubleshooting technical issues.

2.4 Constraints

- The portal must be accessible via web browsers.
- The database should maintain student privacy and data security.
- Response time for viewing or submitting assessments should be minimal.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

- **Login Screen:** Allows users (students, teachers, admins) to log in securely.
- **Dashboard:** Displays an overview based on user type (e.g., assignments for students, submission tracking for teachers).
- **Assignment Creation & Submission Pages:** Allows teachers to create assignments and students to submit them.
- **Grading Interface:** Allows teachers to grade assignments, add comments, and provide feedback.
- **Admin Panel:** For university administration to manage users, courses, and allotting course.

3.1.2 Hardware Interfaces

- The IA Portal can be accessed from any device with compatible browser
- The User should have a good internet connection with minimum basic requirement(
• 1GB of RAM, 8GB of Storage)

3.1.3 Software Interfaces

- **Database:** MongoDB is used to store data on users, assignments, submissions, and grades.
- **Web Server:** Express is used for handling requests and performing operations.

- **Authentication:** JWT (JSON Web Tokens) for securing user sessions.

3.1.4 Communication Interfaces

- **HTTP/HTTPS Protocols:** Used for secure and fast communication between the client and server.
- **SMTP:** Configured to send notifications about deadlines, submissions, and feedback.

3.2 Functional Requirements

3.2.1 Mode 1: Student Mode

- Students should be able to view a list of assignments for their enrolled courses.
- Students should be able to submit assignments by uploading files or entering text responses.
- Students should be able to view submission status, feedback, and grades.
- Students should be able to view announcements or updates from teachers.

3.2.2 Mode 2: Teacher Mode

- Teachers should be able to create assignments by specifying details like title, deadline, max marks, and additional resources.
- Teachers should be able to view submissions from students and mark them as graded or pending.
- Teachers should be able to assign grades and feedback to each submission.
- Teachers should be able to download a report of assignment submissions and grades.

3.2.3 Mode 3: Admin Mode (University)

- Admins should be able to manage user accounts, course (add, delete, or modify student and teacher accounts).
- Admins should be able to oversee course enrollments, including adding or removing students from courses.

3.3 Performance Requirements

- **Load Handling:** The system should support up to 100 concurrent users without a noticeable decrease in performance.
- **Response Time:** Standard actions (viewing an assignment, submitting work) should be completed within 2 seconds.
- **Data Retrieval:** Loading an assignment list or submission history should take no more than 5 seconds for any user.

3.4 Design Constraints

- **Browser Compatibility:** The portal should work on all major browsers (Chrome, Firefox, Edge, Safari).
- **Database Choice:** MongoDB is preferred for handling assignment, submission, and user data.
- **Security Standards:** Compliance with data privacy regulations, ensuring secure handling of student and assignment data.

3.5 Attributes

- **Reliability:** The system should maintain 99.9% uptime, with regular backups to prevent data loss.
- **Maintainability:** The codebase should be modular, enabling updates or fixes with minimal disruption.
- **Usability:** The portal should have an intuitive design, with clear navigation and accessible help documentation.
- **Scalability:** The system should be able to expand to accommodate more users or courses with minimal configuration changes.

3.6 Other Requirements

- **Backup and Disaster Recovery:** Weekly automated backups of the database to prevent data loss.
- **User Support:** An integrated help desk or support system for troubleshooting common issues.
- **Audit Logs:** Maintain logs of all user actions for monitoring and compliance purposes.
- **Accessibility:** Compliance with web accessibility standards (e.g., WCAG) to support users with

Function Point Calculation

Information domain values are defined in the following manner:

- External Inputs (EIs): Data or control inputs from users or other systems that update internal files.
- External Outputs (EOs): Derived information sent to users like reports or screens.
- External Inquiries (EQs): Online inputs that trigger immediate responses, often from internal files.
- Internal Logical Files (ILFs): Grouped data stored within the application.
- External Interface Files (EIFs): External data groups used by the application but not stored within it

Size Estimation of project

- Total external inputs = 7
- Total external outputs = 3
- Total logical internal files = 1
- Total external inquiries = 6
- Total external interface files = 1

Measurement parameter	Weighting factor		
	Simple	Average	Complex
Number of user inputs	3	4	6
Number of user outputs	4	5	7
Number of user inquiries	3	4	6
Number of files	7	10	15
Number of external interfaces	5	7	10

$$\text{Function Point} = \text{UFP} \times \text{VAF} = \text{Count Total} * (0.65 + 0.01 * \sum f_i)$$

(Assuming Simple weighting factor for user input, outputs and inquiries and Average Complexity for internal files and external interface)

Computing Unadjusted Function Point

$$\text{UFP(Total)} = \{7 \times 3\} + \{3 \times 4\} + \{1 \times 10\} + \{6 \times 3\} + \{1 \times 7\} = 59$$

Computing Variable Adjusting Factor

1. **Data Communications** – 3
2. **Distributed Data Processing** - 2
3. **Performance** - 4
4. **Heavily Used Configuration** - 3
5. **Transaction Rate** - 3
6. **Online Data Entry** - 4
7. **End-User Efficiency** - 3
8. **Online Update** - 4
9. **Complex Processing** - 3
10. **Reusability** - 2
11. **Installation Ease** - 1
12. **Operational Ease** - 3
13. **Multiple Sites** - 2
14. **Facilitate Change** – 3

$$0.65 + (0.01 \times 40) = 1.05$$

Adjusted Factor Point is $= 59 \times 1.05$

$$= 61.95$$

Hence

Calculated Function Point is 61.95

USE CASES

Use cases specify the functionality of a system by specifying the behavior of the system, captured as interactions of the users with the system.

1 Creating Teacher,Student,Course&Allotting Course

Primary Actor:University

Precondition:Adminstrator is logged in

Main Flow:(Teacher and Student Creation)

1. Admin login
2. Go to teacher or student creation
3. Enter Details
4. Submit The Details

Main Success:

- Teacher,Student is created and they receive a email with their Unique ID

Extension Scenario:

- if admin enter wrong password it forward to login page
- Admin misfill the field or the same email id is entered then the system popup with valid error reason

(Course Creation)

1. Admin Login

2. Go to Course Section
3. Enter Course ID and Course Name
4. Submit

Main Success:

- Course is Created
- Course is previously Created an error occurred

(Allotting Course)

1. Admin Login
2. Go To Course Allot Section
3. Enter User ID ,Role Or Fetch to see previous Course
4. Add or Modify Course
5. Submit

2. Teacher Login&Creating Assignment

Main Actor:Teacher

Precondition:Teacher Must Be Logged In

Main Flow

1. The teacher enters credentials and successfully logs into the IA portal.
2. The teacher accesses the assignment section from their dashboard.
3. The teacher clicks on "Create Assignment" to start the assignment creation process.
4. The teacher fills in required fields, including
 - Assignment Name
 - Subject Code
 - Assignment Description
 - Deadline (due date for submission)
 - Maximum Marks

5. If the assignment requires additional resources, the teacher can upload relevant files (e.g., PDFs, Word documents, images).
6. The teacher clicks "Upload" to attach the file(s) to the assignment.
7. The teacher clicks "Save" to publish the assignment and system confirms the assignment is created and made available for students to view and submit.
8. The teacher clicks on a student's submission to view the uploaded file and other submission details.
9. The teacher enters a grade (score out of maximum marks) and any feedback comments for the student.
10. The teacher clicks "Submit" to save the grade and feedback.

Extension Scenario:

- If the teacher enters incorrect credentials, the system displays an error message and requests re-entry.
- If there's a system error during creation (e.g., database error), the system displays an error message and prompts the teacher to retry.

3.Student - View and Submit Assignment

Main Actor:Student

Precondition

- The student must be registered in the system and logged in.
- The student is enrolled in courses that have active assignments.

MAIN FLOW:

1. The student enters their credentials and successfully logs into the IA portal.
2. The student goes to the "Assignments" section from their dashboard, which lists all assignments for the enrolled courses.
3. The student clicks on a specific assignment to view details, including:
 - Assignment Name
 - Subject Code
 - Description/Instructions
 - Deadline
 - Maximum Marks
 - Any attached files or additional resources uploaded by the teacher

4. If the teacher has attached any files, the student can download them for reference.
5. The student completes the assignment based on the provided instructions.
6. The student clicks "Submit" and uploads their completed work (e.g., PDF, Word document, image).
7. The student may add any additional comments for the teacher if needed.
8. The system confirms that the assignment has been successfully submitted.
9. After the teacher grades the assignment, the student can return to the assignment to view feedback and their grade.

Extension Scenario:

- The system either restricts submission (if late submissions are not allowed) or flags the submission as late.
- The student is notified of any penalties or consequences for late submission.
- If there's an error while uploading the file, the system displays an error message and prompts the student to retry.
- If the file size exceeds the allowed limit, the system notifies the student to reduce the file size before uploading.